Network Monitoring and Analytics for Data-Driven Observability

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Initiatives:IT Operations and Cloud Management for Technical Professionals

Existing device-centric network monitoring does not scale or provide sufficient visibility for modern applications. To correlate network availability and performance with application health, I&O technical professionals must rely on analytics and AIOps to classify and prioritize network telemetry.

More on This Topic

This is part of an in-depth collection of research. See the collection:

- What I&O Leaders Need to Know About Observability

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Overview

Key Findings

- Network monitoring vendors are continuing to pivot to unified platforms that process a combination of logs, metrics, flows, packet capture and configuration data. In the past, enterprise network monitoring used disparate device- and protocol-centric tools.
- Specialized network monitoring functionality has not yet converged into the leading observability platforms, leaving a gap for network operations teams and necessitating the use of separate tools or platforms for network performance monitoring (NPM).
- The emergence of integrations with telemetry pipelines has opened up new opportunities to generate increasingly accessible advanced analytics and to leverage AIOps features for automated network observability and management.
- Monitoring tools for software-defined infrastructure, such as those available from cloud providers, demonstrate the paradigm shift from stack- or silo-based to unified and data-driven.
- AIOps features increasingly support automated root cause analysis as networks increase in size and complexity.

Recommendations

- Prioritize platform-based solutions that have the ability to analyze, augment and aggregate data. Assess the current vendor capabilities in terms of enrichment, data exploration and AI-assisted analysis.
- Consolidate monitoring tool portfolios into a telemetry pipeline that supports network-automated problem identification and resolution. Prioritize the data and capabilities that mitigate blind spots or provide the most value.
- Perform real-time data analytics for applications and network diagnostics by leveraging technology such as low-latency flow monitoring, streaming telemetry and network-centric digital experience monitoring.
- Supplement end-to-end and infrastructure data by using path analytics and synthetics to gain visibility into those portions of the network that I&O professionals traverse, but don't control, such as the internet.

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Problem Statement

Traditional monitoring tools are mostly reactive and centered on the health and performance of individual network elements. Moreover, the network tools themselves are siloed by their data capture mechanisms. IT infrastructure monitoring (ITIM) and NPM tools are evolving to a more networkwide and service-centric approach, and Gartner customers are using these tools more proactively. Cloud and digital-era network monitoring need to correlate individual element information not only from the network as a whole but also from the applications that rely on it. The agility that digital business, cloud and dynamic infrastructure demand requires lower mean time to repair (MTTR) and mean time to detect (MTTD).

For years, organizations used many tools, forcing employees to toggle between multiple nonintegrated systems and, often, a lack of end-to-end data collection. This approach is no longer necessary or advisable. New techniques and tools coming from machine learning (ML) and data analytics can assist network monitoring teams in gaining insight, and they can guide or automate remediation. There are tools that can synthesize all these different telemetry streams and provide an informed analysis.

Three types of integration that can mitigate the impact of tool proliferation are:

- **Front-end integration,** where the tool's user interface provides direct links to other data types, such as Simple Network Management Protocol (SNMP) utilization of an interface linking to NetFlow or Grafana's ability to visually present telemetry from multiple, disparate repositories.
- **Back-end integration,** where data from multiple monitoring tools is aggregated into a single repository or analytics solution, sometimes called a bridge database or data lake. Users can operate on the combined dataset by using a third-party data visualization tool such as Microsoft Power BI or Tableau.
- **Event aggregation,** such as with event intelligence solutions. This is where events from multiple monitoring solutions are consolidated into a single console, with onboard analytics usually providing additional benefits such as cause analysis and anomaly detection.

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Ultimately, these new techniques will reduce the time network operations will spend chasing symptoms and afford more time to solve problems. The siloed-tool approach does not easily accommodate the application of analytics, ML and data science techniques in network monitoring. It may be easier to adopt a data-centric network monitoring approach for workloads residing in an infrastructure as a service (IaaS) cloud provider. Cloud-native monitoring tools (such as Amazon CloudWatch, Microsoft Azure Monitor or Google Cloud Operations) are less piecemeal and can observe all infrastructure and application layers.

However, Gartner customers often find cloud tools lacking in features and visibility, and they don't integrate well with on-premises tools or across clouds. This lack of physical layer visibility also applies to traffic traversing networks not owned or subjected to carrier SLAs such as internet-bound traffic. Network operations teams are forced to operate differently, depending on whether the networks they are managing are in the cloud or on-premises.

This document guides organizations in evolving their network monitoring practices, addressing modern workload performance, visibility and observability requirements. It addresses the question:

> **How can I transform my network monitoring technology and practices to be more proactive and application-, user- and business-centric?**

The Gartner Approach

To augment network visibility and act based on the operator's network situational awareness, Gartner proposes a data-driven network monitoring framework. The framework enables organizations to leverage advances in analytics, telemetry pipelines, and event intelligence technology. The framework is domain-based, addressing each domain individually. However, the domains meet in the network telemetry pipeline.

Organizations following this framework will be able to shift their attention from network protocols for gathering data to the data itself. The gathering method becomes a means to an end. However, the framework addresses emerging techniques to collect data, such as streaming telemetry. Table 1 maps data gathering methods to data types.

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Monitoring, by definition, suggests that you know what you're looking for. We call those "known knowns" and "known unknowns." But what happens when you don't know what you're looking for? The term "observability," borrowed from control theory, has entered the IT monitoring lexicon to describe a holistic, data-centric approach that emphasizes an exploratory capability and enables identification of "unknown unknowns." Monitoring emphasizes externally visible signals, while observability relies upon detailed representations of a system's internal state. Network monitoring data takes many forms and can represent both perspectives, depending on what is being investigated.

Table 1: Data Type and Collection Methods

(Enlarged table in Appendix)

Domain	Data type	Purpose	How to collect the data
1	Log/Event Data	Awareness of changes and events to network devices and cloud components providing network services	SNMP traps, syslog, VPC Flow Logs
2	Metric Data	Collecting performance and use over time	SNMP polling, streaming telemetry, API retrieval, synthetic transactions, endpoint agents
3	Flow Data	Collecting metadata on end-to-end conversations in the network	NetFlow, sFlow and IPFIX
4	Packet Data	Collecting full conversation content for payload diagnostics and security use cases	Port mirroring, port spanning technologies, network packet brokers, taps
5	Configuration Data	Capturing device configuration change management	Screen scraping, NETCONF/yang, API, Ansible, network source of truth (NSoT) repositories
6	Forwarding, Route and Path Data	Optimizing traffic flow between endpoints	Network trace, API, polling, passive route collection

Source: Gartner

The framework will provide three-dimensional guidance for every domain while addressing:

- How to collect the data
- Why the data should be collected
- How to use the data

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The collected data will enter into a telemetry pipeline sharing data from multiple IT silos and use various tools to ingest, normalize and correlate the data. The telemetry pipeline, whose role is to unify and analyze the data, is addressed in Domain 7 in Figure 1.

Ultimately, the goal is to leverage data-centric technology innovations that enable:

- Transforming data into insights
- Transforming insights into actions

The Guidance Framework

Following a brief introduction to the goals of network monitoring, the Gartner data-centric network management framework has three main components:

- The prework component addresses the information to meet the goals of network monitoring. Prework includes a discussion of data gathering, generation techniques and the time aspects of data. We call this information the “golden signals” of networking.
- The second component covers each of the six data domains to collect (as shown in Table 1). Each domain will itself have four sections: the domain definition, how to collect the data, how to use the data or what the data is good for, and domain-specific guidance.
- The third component is a section on the telemetry pipeline, which will guide the reader on how to leverage network telemetry pipeline and AIOps features to gain insight and act on network monitoring information.

Figure 1 illustrates the framework. The first six domains address the various data to collect. Domain 7 represents the telemetry pipeline section.

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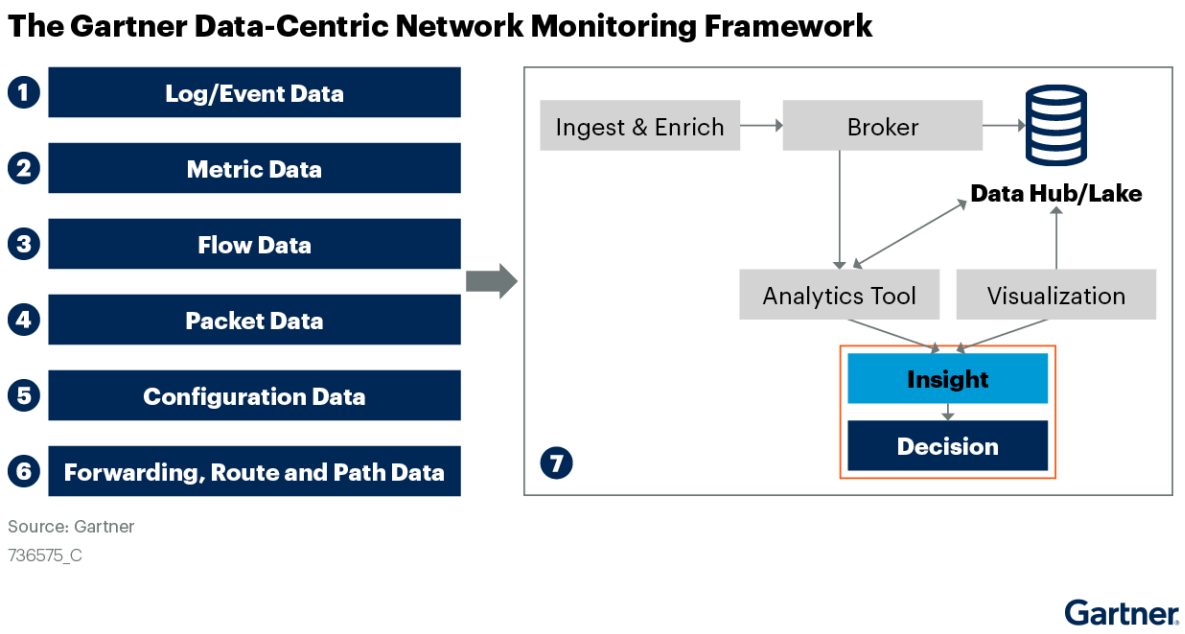
Figure 1. The Gartner Data-Centric Network Monitoring Framework



As analytics tools mature, network monitoring data should merge with monitoring data from other infrastructure monitoring or observability platforms as well as application and business data that leverage the telemetry pipeline.

> Part of network monitoring's shift to data and analytics commoditizes the collection mechanisms and pushes tool differentiation toward optimizing collection and storage, accelerating insights and automating problem resolution.

Prework

Determining what to monitor is critical to scoping monitoring tools and platforms, [1] especially because collecting too much network data is possible. An overabundance of data will tax the collection storage system. It will also place unnecessary demands on technology resources and personnel, and can cause unnecessary alerts. Ultimately, too much data can obscure the root cause and slow down problem resolution.

The Goals of Network Monitoring

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An understanding of the goals of network monitoring helps rationalize and set the tone for what to monitor. The collected data should meet one of the following three goals:

- Gain the visibility to ensure adequate network support of applications, based on:
  - Quality of user experience
  - Availability
  - Performance and utilization, including link utilization, packet loss, latency and jitter
- Ensure early detection of availability and performance problems
- Provide problem resolution assistance by pinpointing the source of a problem, offering insights into the root cause and possibly suggesting actionable solutions.

Blindly gathering data for analytics purposes is tempting, but organizations need to understand how the monitoring data they collect maps back to their network monitoring goals. This remains true when using analytics, even when collecting data strictly for further analytics purposes. Refer to Guidance Framework for Deploying Centralized Log Monitoring for more details on implementing collection standards.

Organizations should focus their network monitoring efforts on gaining network visibility, ensuring early detection of application performance and availability problems, and providing network resolution assistance.

The bulk of this research concentrates on data types and sources. Beyond the data source, we categorize data as passive or active as well as real time or historical.

Passively Gathered Data Versus Actively Generated Data

Passive data collection observes existing network traffic, either by using instrumentation that's built into devices or by using special hardware that watches network traffic. Examples of passive data collections include SNMP, flow-based data, streaming telemetry, packet captures, log data, and forwarding/routing information.

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Active data collection inserts synthetic test traffic into the system. Cisco's IP SLA, which simulates application traffic and generates metrics, is an example. In this case, fleets of agents distributed across the internet generate traffic to various SaaS providers, as offered by vendors such as Catchpoint, ThousandEyes and Obkio. It is difficult to collect device-based data in a network you don't control or own. So, you need to monitor the networks you don't own from the outside; you can't get device-centric or passive data.

Digital experience monitoring (DEM) supports both the collection of telemetry about user activity (sometimes called real-user monitoring or RUM) and the initiation of synthetic traffic to establish performance baselines or determine availability when there are no "real users."

Gain additional visibility for traffic traversing the internet and augment flow-based data with network synthetic methods.

Historical and Real-Time Data

Supplement historical data analysis with real-time data analysis. Acting on network conditions requires real-time data to either suggest or automatically trigger a change in the network environment. Advances in data science and analytics enable new types of real-time data solutions, such as the ones offered by Pico's Corvil Analytics, ExtraHop and Kentik.

Future autonomous, self-healing networks will heavily depend on real-time data processing and advanced analytics.

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Historical and real-time data work well with each other. Incident management, which aims to restore service during an outage, will benefit from real-time data. Problem management, incident management's counterpart, which aims to find the cause of an outage, leverages historical data. Trending and capacity management is another use of historical data. Event intelligence solutions enhance IT monitoring capabilities, including, but not limited to, anomaly detection, automated root cause analysis, predictive analytics and augmented remediation. Event intelligence solutions provide tactical visibility for monitoring where multiple different datasets are coming together, enhancing efficiency. See Infographic: Using AIOps for Managing Operational Telemetry for additional details.

Domain 1: Log and Event Data

Log and events data evolved from collecting logs in circular buffers no bigger than a few KB in size to centralized, log-aggregating databases executing real-time analytics and alerting. Log and machine data also include SNMP traps, which are unsolicited SNMP messages based on events or thresholds sent to a management console. As the name implies, log and events data is generated by the system logging facility or by a user-defined threshold; therefore, it is possible only in networks you own or control.

1.1 How to Collect the Data

To collect this data, enable SNMP traps on your devices or configure your syslog-based systems to send logs to a syslog server. Log data is typically collected by a syslog (or a variant such as rsyslog or syslog-ng) and sent to a log repository on the network. There are four major components in a network logging architecture, as Figure 2 illustrates:

- The collector component that gathers and ships logs from the network devices, on the left in Figure 2
- The transform and aggregate layer that routes data and can duplicate it to multiple sources as well as transform and normalize it
- The log repository that ingests and stores logs (Log Server in Figure 2)
- The analytics and visualization engine (UI/API in Figure 2)

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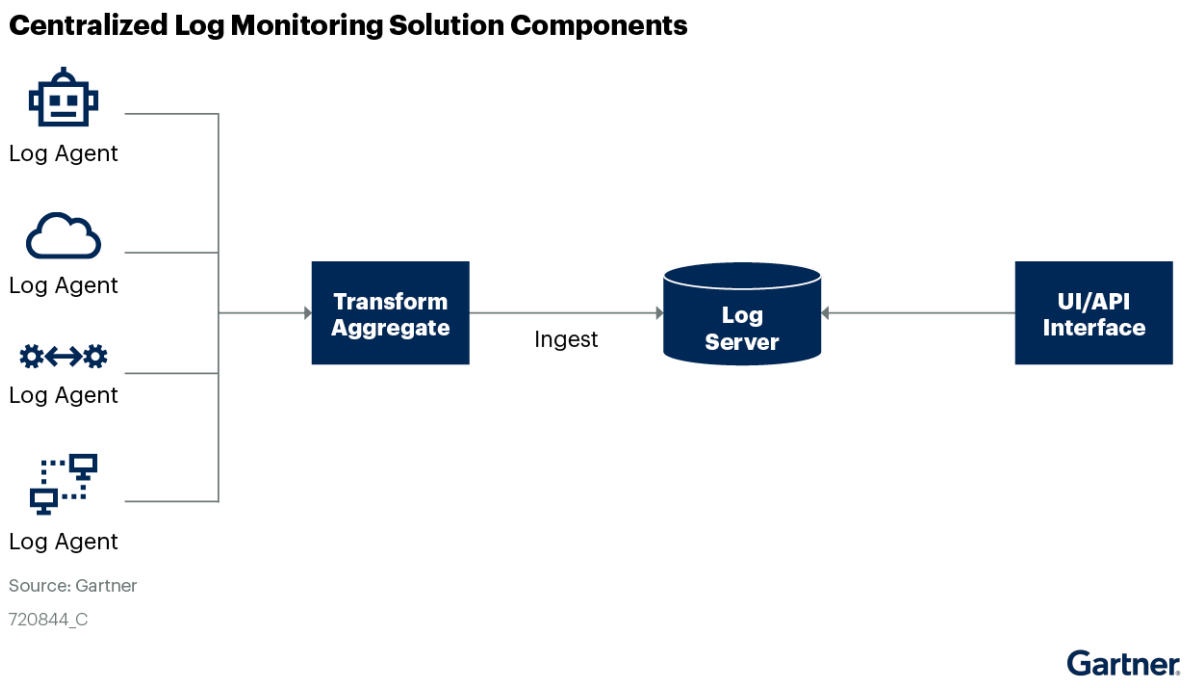
Figure 2. Centralized Log Monitoring Solution Components



For example, the open-source Elastic Stack consists of Beats for log collection, Logstash for log routing and transformation, Elasticsearch for indexed log storage, and Kibana for log analytics and visualization. Modern logging and analytics systems such as Elastic Stack, Splunk and Logz.io follow this architecture.

The syslog-based system has multiple facilities, analogous to channels. Each syslog facility has multiple levels of granularity based on severity. See IETF's RFC 5424 for details.

SNMP traps asynchronously send unrequested signals, called "traps," to the management platform. SNMP uses UDP port 162 for these trap messages, which contain variables indicating the trap type (such as "cold start," which normally means that the SNMP agent has rebooted) and associated variables. Traps are commonly sent to indicate cold and warm starts and link state changes. They're also used to transmit alarms, such as temperature and power supply problems. Some advanced management information bases (MIBs) can send alarms because thresholds have been violated or because of other programmable conditions.

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Gartner customers typically send SNMP traps and log events to network fault management (NFM) system tools and event correlation and analysis (ECA) tools for processing. Modern log and event analytics systems can also receive and process SNMP traps.

1.2 How to Use the Data

Log data enables visibility of what happened — in near real time — in the case of SNMP traps. Log and machine data are mostly used for:

- **Forensics:** Logging allows you to investigate what happened to your network as well as when and why. Analytics systems leveraging logging data stores, as in Figure 2, can discover patterns and create actionable insights. Eventually, analytics tools will take advantage of the log data to identify issues both reactively and proactively, using automation to resolve those issues.
- **Event notification:** SNMP traps are used to set up real-time alerts, such as interface down, failed hardware component (fan is out) or high link utilization.

Log and machine data offer a retroactive view of the network. Additionally, the actionable insights produced by analytical system processing logs are typically device-based as opposed to flow-based because the data itself is device-based. Log and machine data must be an essential component of every organization monitoring strategy. However, log and machine data is not sufficient to obtain network visibility at the flow level or for early detection of events or anomalies.

1.3 Guidance

Gartner recommends that technical professionals:

- Collect log and event data in a centralized repository with search and analytics capabilities, as Figure 2 illustrates.
- Consolidate your network and operational logging.
- Evaluate SaaS-based log management systems to scale and simplify operations.
- Apply metadata and tags to your log data to provide context for search in indexing. The tagging typically occurs in the transform phase and can be done automatically in many circumstances.

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- Consider leveraging telemetry pipeline products such as Cribl or Mezmo for analysis, enrichment and routing of logs in real time.
- Establish standards for alerting, and consider leveraging an incident response automation system, such as Atlassian (Opsgenie), xMatters or PagerDuty. See ITSM Best Practices: Automating Incident Management for more details.

Domain 2: Metrics Data

Metrics data is the most common type of quantitative performance data collected and represented over time, and it is present in almost every kind of monitoring. A single instance of a metric — sometimes called a data point — includes three required attributes:

- Name: The identifier for the metric
- Time stamp: The date/time associated with this data point
- Value: The quantity associated with this data point

By collecting and indexing data points throughout time, we arrive at the term "time series." The terms metrics data and time series data are used synonymously in this research.

Today, most metrics include a fourth field, often called "labels," that gives a set of key-value pairs to deliver additional context about the data. If a particular label is indexed, the label's value represents an additional time series with which the data point can be displayed or analyzed. An unindexed label value can be used as context to annotate a visualization, but otherwise has little impact on the underlying store. For more information, see Monitoring and Observability for Infrastructure and Applications.

The nature of time series data and the impact of cardinality on read and write performance make general-purpose databases poor repositories for metrics data. Table 2 includes examples of solutions that provide optimized storage for time series data.

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Table 2: Representative Time Series Data Management Solutions

Solution	Vendor	Open Source	Delivery Model
Grafana Mimir	Grafana Labs	Yes	SaaS, self-hosted, open-source software (OSS)
InfluxDB	InfluxData	Yes	SaaS, self-hosted, OSS
M3	Chronosphere	Yes	SaaS, OSS
VictoriaMetrics	VictoriaMetrics	Yes	SaaS, self-hosted, OSS
TimescaleDB	Timescale	Yes	SaaS, OSS
Amazon Timestream	Amazon Web Services	No	SaaS
Time Series Insights	Microsoft Azure	No	SaaS
Timeseries Insights API	Google Cloud	No	SaaS

Source: Gartner

2.1 How to Collect the Data

It is likely that metrics are available from every (or almost every) device across your network infrastructure. Existing network monitoring tools may already be collecting those metrics. Common collection vectors include:

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- "Pull" mechanisms:
  - SNMP (GET)
  - API (often REST/JavaScript Object Notation [JSON])
  - Prometheus (OpenMetrics)
- "Push" mechanisms:
  - Streaming telemetry
  - Software agents, such as collectd and InfluxData's Telegraf.

Pull refers to an active, polling process that requires the monitoring data collection to be originated by the collector. Push, on the other hand, is passive on the part of the collector, with the metrics being sent by the agent periodically or via a stream.

The mechanisms are more malleable than the above list suggests, allowing for combinations — for example, both collectd and Telegraf can themselves "pull" data using SNMP that can then be "pushed" to a collection repository.

See Reference Architecture Brief: Software Observability for additional guidance on specific components that can be used to send, receive and store metrics at scale.

Streaming Telemetry

At the speed of today's networks, things happen very quickly. The common polling intervals when using SNMP retrieval (and similar mechanisms) are usually too long and do not provide the accuracy, time series granularity or flexibility necessary to make decisions in near real time. Vendors, such as Arista Networks, Cisco and Juniper Networks, and initiatives, such as OpenConfig, enable streaming telemetry technology that uses a subscription mechanism. The streaming analytics mechanism allows an interested management entity to "subscribe" to the data source and receive operational statistics, events and configuration changes from network devices automatically and almost continuously. Most streaming telemetry mechanisms leverage gRPC, a high-performance, open-source universal RPC framework, based on Google protocol buffers. However, streaming telemetry has not been widely adopted due to vendor-specific constraints and general hesitation by network operations teams to learn and implement it.

2.2 How to Use the Data

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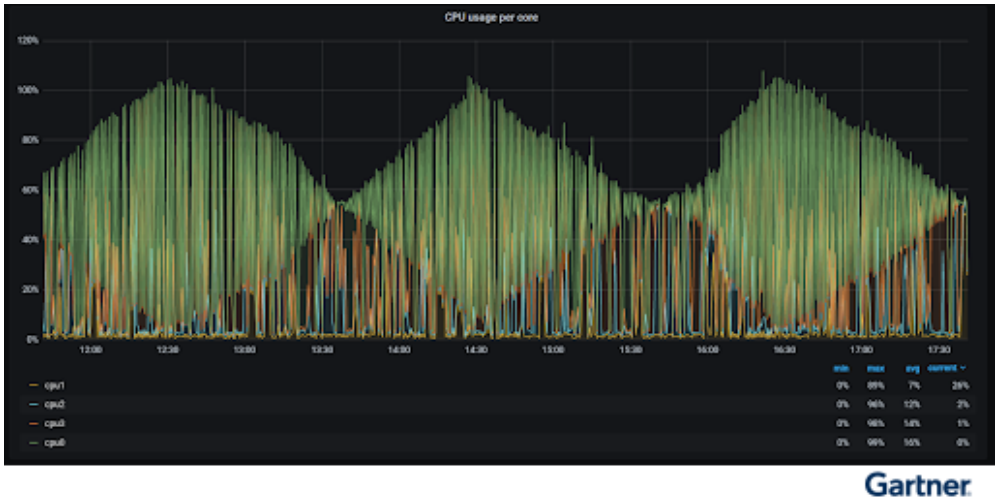
Metrics data is the backbone of monitoring. Utilization and throughput are some of the types of signals, usually managed as time series data. I&O processes that leverage metrics include failure detection and notification, as well as infrastructure, bandwidth and performance management. Figure 3 illustrates the kind of visualization commonly backed by time series data.

Figure 3. Visualizing Metrics Data



Source: Gartner

2.3 Guidance

Gartner recommends that technical professionals:

- Identify the key metrics that indicate the health and performance of the network, and use these metrics to generate alerts. Interface utilization, latency between network devices and error rate are examples. Any numeric data point that can be collected and contextualized over time can be a metric.
- Eschew static thresholds in favor of algorithms or statistical analysis to identify deviations from normal.
- Separate current raw data from historical aggregated or averaged data in their metrics storage. Timely, accurate analytics requires access to raw data, while historical data may still be valuable for longer-term investigation of security issues. Most modern time series stores include built-in aggregation capabilities that automatically aggregate and expire metrics over a series of relevant time horizons. Ensure that the raw, real-time data store is of sufficient size to meet specific needs.

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- Balance the utility of metrics data with the limited context it can carry. Labels ensure that the data shows up in the correct time series but doesn't include the richness of the event, packet and flow domains, as discussed in other sections of this research.
- Implement a visibility strategy that prioritizes streaming telemetry but leverages SNMP where logical.

Domain 3: Flow Data

Flow data adds application context to network monitoring by collecting metadata on conversations. Flow data groups unidirectional packets that are sharing the same properties, such as source and destination IP addresses and protocol. Both network performance and security tools leverage flow data. Metadata about flow keeps only the critical information and generates less data than packet data; thus, it can be stored for much longer time periods. It is also possible to sample collection, by flow or by packet, to reduce data volume, depending on the flow solution.

Earlier forms of flow data protocols, such as Cisco NetFlow, introduced in 1996, grouped packets based on the unidirectional septuple attributes:

- Ingress interface
- Source IP
- Destination IP
- IP protocol
- Source port
- Destination port
- IP type of service (TOS)

Although Cisco NetFlow is the predominant variant, today there are additional formats based on the technology. This includes formats such as IPFIX, an IETF standard extended from NetFlow v9, Juniper Junos Traffic Vision (previously known as Jflow) and Huawei NetStream. sFlow, a Layer 2 technology, can only sample packets. Moreover, sFlow does not create flow on the network device. The network device sends packet data to a collector, which creates the flows. Implementing sFlow is more processor-intensive than NetFlow-based technology, and it contains Layer 2 information, such as VLAN ID, making it popular for Layer 2 devices.

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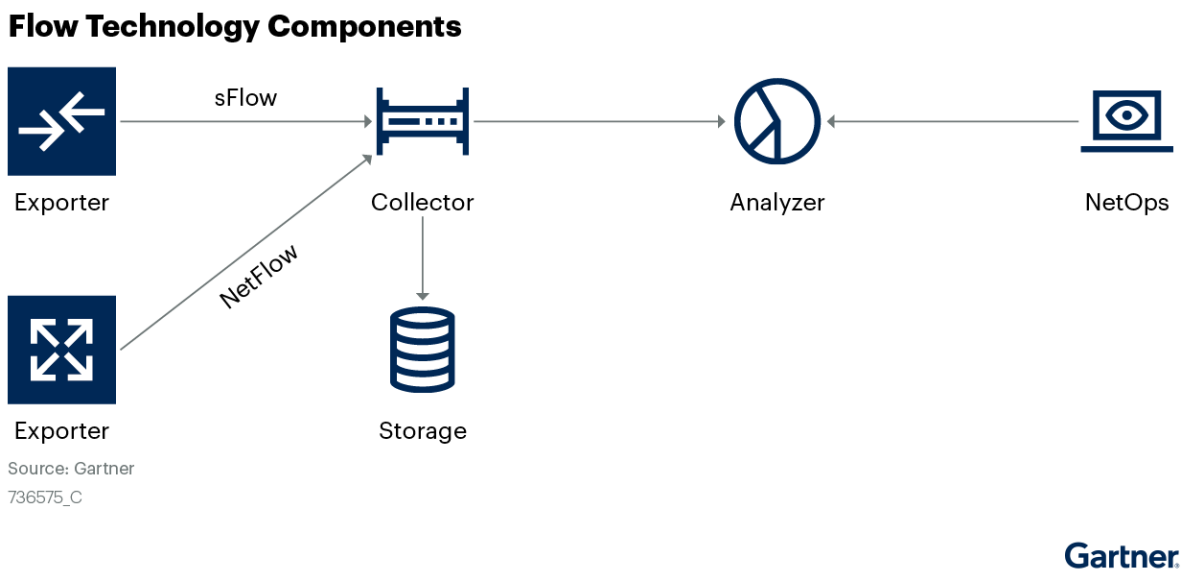
3.1 How to Collect the Data

Network devices that support flow technology send flow data to central collectors, where traffic analysis occurs, as displayed in Figure 4. The network device plays the exporter role. Additionally, network packet broker (NPB) vendors like Arista, Gigamon and Keysight can also generate flow information by creating flow metadata in parallel with processing the data traffic stream, thus becoming exporters.

Figure 4. Flow Technology Components



Typically, a device supports either sFlow or a NetFlow variant, forcing some organizations to use both sFlow and a variant of NetFlow. Flow technology storage and processing can incur significant hardware and maintenance costs. Organizations should evaluate SaaS-delivered offerings, such as Kentik, Auvik and VMware vRealize Network Insight (vRNI), to mitigate the hardware costs.

3.2 How to Use the Data

The application and connectivity context provided by flow-based technology enable organizations to:

- Identify top talkers on a link
- Understand the traffic profile
- Optimize internet peering relationships
- Detect unauthorized traffic

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- Validate quality of service (QoS) parameters
- Establish the presence of network anomalies

3.3 Guidance

Gartner recommends that technical professionals:

- Collect flow data for all Layer 2 network devices at the edge of their networks, including the data center edge and the WAN edge, and send data to a centralized flow analyzer. Flow data is useful for getting visibility into a branch where packet collection and analysis is hard to perform.
- Use analytics tools that can correlate flow and types of data, such as packet data or routing information.
- Evaluate SaaS-based flow analysis tools. Organizations that adopted hybrid cloud technologies should prioritize SaaS-based tools.

Domain 4: Packet Data

Packet data, the capture of raw network packets traversing a section of the network, is the ultimate source of network truth. It differs from flow data technology, which creates metadata based on the packet headers. Packet data contains both the header and payload of each packet.

Packet data is expensive to capture and retain because it will consume a large amount of storage space and generally must be captured and written to disk at wire speed. Because of the potential hardware costs, organizations must restrict packet data technology usage and retention to specific areas of their network, such as the “demilitarized zone” (DMZ) or around their critical applications.

4.1 How to Collect the Data

Organizations can use a variety of extraction methods to collect packet data. The collection method varies depending on whether the network is physical or virtual. Moreover, if the network is virtual, the collection method depends on who owns the hypervisor. In private data centers, where I&O controls the hypervisor and physical network, it’s possible to mirror virtual machine (VM) traffic based on the virtual network interface card (vNIC) or run the vNIC in promiscuous mode and install a virtual tap. Amazon Virtual Private Cloud (Amazon VPC), with Traffic Mirroring, and Google Cloud Platform (GCP) both offer packet capture services.

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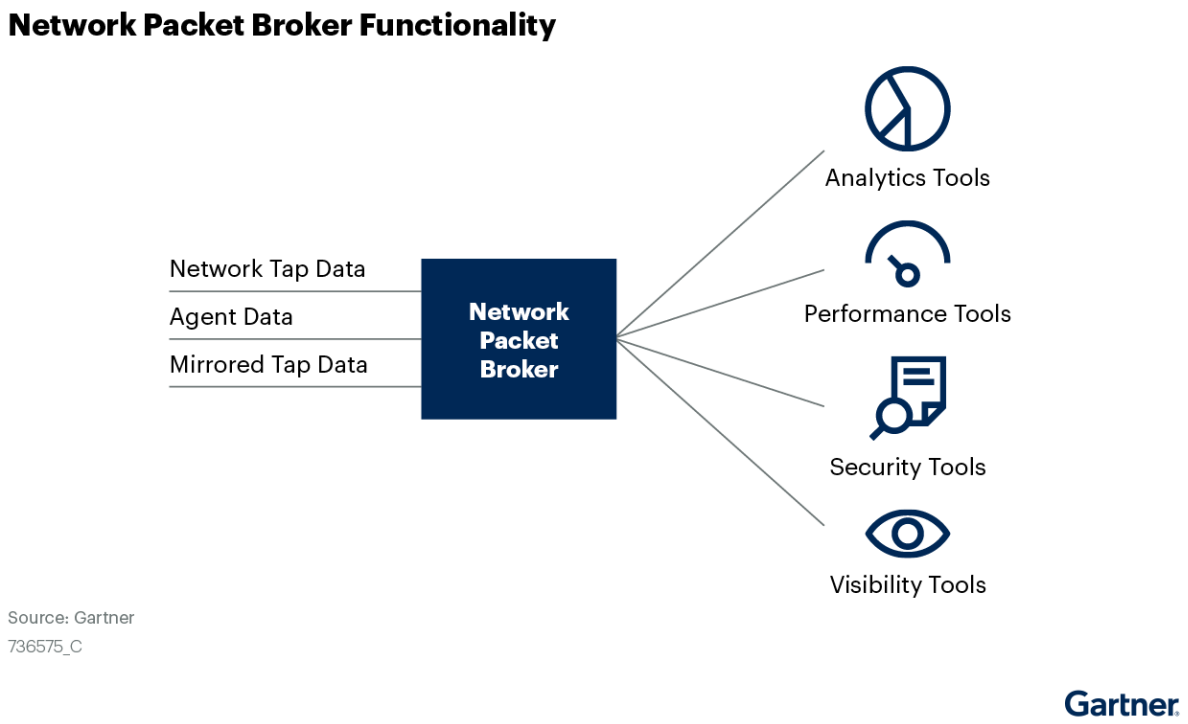
However, Azure's packet capture service, Virtual Network TAP, is in preview mode. Organizations that don't control the hypervisor, such as in a public setting, might be forced to deploy agents in every VM in which they want to collect the traffic, unless the cloud provider offers a virtual network tap service. Vendors such as Gigamon offer either VM or container-based agent deployment that is compatible with all major cloud service providers.

Because various tools, including security tools such as intrusion detection systems, leverage packet data, organizations should deploy NPBs to enable efficient distribution of packet data across multiple different toolsets, as Figure 5 illustrates.

Figure 5. Network Packet Broker Functionality



Modern NPBs can also provide:

- Packet stream aggregation to redirect multiple data inputs into one stream
- Decryption, provided the NPB contains access to the appropriate encryption keys
- Data tagging to classify traffic and simplify search and storage
- Filtering and splicing of data (on a per-tool basis) to discard irrelevant traffic or to send only a subset of the packet payload

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- Data masking and obfuscation to comply with data privacy policies
- Generation and forwarding of flow data

To scale collection capability, large network operators build parallel networks of NPB devices. The NPB vendors provide central management of the packet broker complex, allowing operators to configure and administer all filtering, aggregation, disaggregation and deduplication workflows from a single location.

4.2 How to Use the Data

Packet data has complete visibility into the network and applications running on it, but network operators need specialized tools to analyze packet data. Eyeballing a packet capture file using a tool such as Wireshark can help to troubleshoot a single application issue. However, it will not provide any predictive analysis or generate insights on the state of a highly utilized, complex network. The main benefits of packet capture are using decodes to understand application behavior, looking at time stamps to understand end-to-end performance and looking into the payload to get business intelligence (BI) information. Replaying application transactions for troubleshooting is another packet data use case.

The P4 data plane programmability language can modify packet headers at the hardware layer while preserving line-rate speeds. For example, emerging analytical and monitoring tools can leverage P4-compliant application-specific integrated circuits (ASICs). These include smart network interface cards (NICs), such as Netronome and AMD Pensando. They can be leveraged to process additional packet header information, such as a time stamp injected by a P4 program.

4.3 Guidance

Gartner recommends that technical professionals:

- Capture packet data for the critical regions of the network, including the DMZ, and where business applications are hosted
- Use NPBs to scale packet capture and allow multiple tools to leverage the captured data
- Prioritize packet-based technology that supports real-time and near-real-time analytics

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- Create a packet data-retention policy by answering the following questions:
  - How long should a full packet be stored?
  - When do you expire the payload and keep the header only? Payload data takes storage space.
  - When should you process and summarize the data, leaving only the headers for analysis?
  - How many hours or days of historical payload data do you need available to sufficiently investigate and issue?

Domain 5: Configuration Data

Configuration data is usually a topic for network automation rather than network monitoring. However, configuration changes in data must be monitored. Most organizations save their network device running configurations and should also monitor and record their network configuration changes. The change configuration deltas should have enough information to roll back to the previous configuration.

Saving and tracking configuration changes have been performed for years by commercial network configuration change management (NCCM) tools and open-source software like Really Awesome New Cisco confIg Differ (RANCID). NCCM tools are offered by vendors like Ignite Technologies (AlterPoint), Auvik, Cisco, Hewlett Packard Enterprises (HPE), Infoblox, ManageEngine, NetBrain and SolarWinds. Most NCCM tools also integrate with ITSM tools such as BMC (Remedy) and ServiceNow. The integration can enforce that only approved changes are pushed to production.

The DevOps and cloud era warrant new, more declarative network configuration technology as “infrastructure as code” principles prevail. Gartner customers are increasingly adopting tools such as Ansible to automate their network. The code itself becomes the source of truth when using infrastructure as code techniques, managed using revision control systems such as Git. See Best Practices for Modernizing Your Enterprise Network for additional guidance on network automation.

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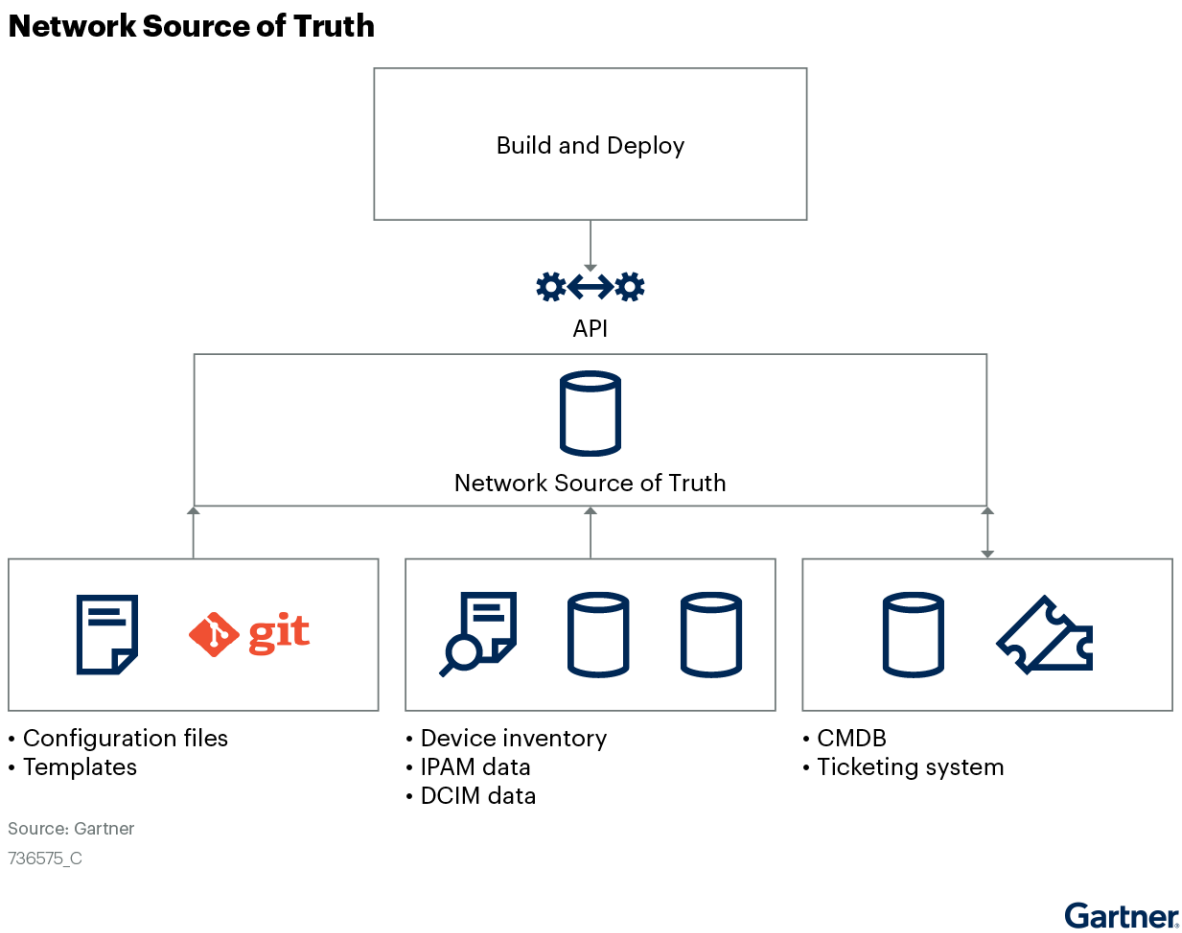
The required information resides in the NSoT, which is made using various data repositories, as seen in Figure 6. The data in the source of truth must itself be available programmatically via API calls. NetBox, an open-source IPAM and DCIM solution initially built for DigitalOcean, is gaining a lot of traction among network organizations adopting infrastructure as code. The vendor Network to Code now offers commercial support for NetBox. NSoT is another open-source solution.

Figure 6. Network Source of Truth



5.1 How to Collect the Data

Historically, organizations used either Trivial File Transfer Protocol (TFTP) transfer or screen-scraping techniques that emulated console access to gather device configuration files. Gathering configuration data has become easier as more and more vendors have added support for streaming configuration data in JSON or XML formats.

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Today, network vendors offer facilities to archive network configuration and support NETCONF, Ansible and Python on their last-generation switches. Multiple Python-based libraries, such as NAPALM, Netmiko, Nornir and Junos PyEZ, are also available. These techniques, together with network discovery systems, are used to build the NSoT.

5.2 How to Use the Data

Configuration change deltas are invaluable data points to troubleshoot problems, as more than 60% of issues are directly caused by network changes. Intent-based network verification tools, such as the ones Juniper, Cisco and Forward Networks offer, as well as projects like Batfish, leverage network configuration in concert with forwarding data to build formal and verifiable models of the network.

By leveraging these tools, it is possible to accurately answer questions such as, “Can these two servers talk to each other if we lose a particular link or network device?” The intent-based verification tools can prove if a given network capability is implemented and meets their operator’s specification. Performing “what-if” analysis based on candidate configuration change is another intent-verification tool use case.

5.3 Guidance

Gartner recommends that technical professionals:

- Collect and centrally store network configuration data. Use modern tools when possible. Only resort to screen scraping when vendors don’t support other methods.
- Establish an NSoT that contains authoritative network configuration information.
- Adopt infrastructure-as-code tools, such as Git, to store configuration data.
- Ensure that the running configuration (in memory) is in sync with starting configuration (in NVRAM or disk).
- Make the configuration data available to other systems.

Domain 6: Forwarding, Routing and Path Data

Forwarding table, routing and path data identifies the path the network packets take. Visibility into sudden changes in routers and switches forwarding tables can help foresee application or network performance behavioral changes. Historical data can also help find the root cause of performance degradation or loss of connectivity.

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Networking literature often uses the term routing information base (RIB) to denote routing tables and forwarding information base (FIB) to denote forwarding tables:

- The forwarding table information determines the interface where a frame or packet should be sent, depending if Layer 2 or Layer 3 is used. Most intent verification systems rely on individual network devices' forwarding tables to build a model of the network.
- Routing information applies to Layer 3 network devices, which typically participate in one or multiple routing protocols. Each routing protocol instance has its own routing table. The network device forwarding table leverages the routing tables to build its forwarding table. In other words, the FIB leverages the RIB to build its table.
- Path information data lists possible network paths traffic takes between two endpoints or networks. Metadata information, such as the router Border Gateway Protocol Autonomous Systems (BGP AS) number, is added to the path. The traceroute utility, which provides the list of routers on the network path between a source and destination point, represents the earliest and simplest example of path data.

6.1 How to Collect the Data

Organizations typically rely on a third-party product that offers alerting and analytics to collect forwarding routing and path data. Intent verification vendors collect forwarding table information, and vendors such as Ciena, SolarWinds and ThousandEyes offer route analytics products. ThousandEyes augments its routing data by collecting data from distributed agents. Getting the forwarding table of a device is similar to getting its configuration file. A data collecting process emulates a console session for older devices and leverages modern methods such as APIs or NETCONF for modern devices. Collecting routing protocol data is typically done by inserting a passive routing participant in the network that records routing updates.

Flow technology like sFlow can also calculate path data because it supports autonomous system information. NetBrain provides context over traceroute path data to assist real-time analysis and troubleshooting.

6.2 How to Use the Data

Organizations use forwarding, routing and path data to:

- Monitor networks they traverse but don't own, such as the internet.

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- Perform intent verification in conjunction with configuration data.
- Alert and log routing and forwarding for incident investigation.

Route analytics tools, such as Packet Design, provide routing state visibility, monitoring and reporting to optimize and troubleshoot services. Path analytics tools such as Entuity SurePath and ThousandEyes collect and visualize paths to ensure traffic takes the best path when multiple paths are available. Path data can also pinpoint traffic not following symmetric paths and other traffic deviation. Asymmetric routing can affect both performance and availability because firewalls typically block asymmetric routes.

6.3 Guidance

Gartner recommends that technical professionals:

- Monitor networks they don't own but traverse, using agents and leveraging path data.
- Monitor and report on route changes and path deviation to troubleshoot services.
- Use forwarding in conjunction with configuration data for intent verification and what-if analysis.

Domain 7: Bringing It All Together With the Telemetry Pipeline

This domain does not represent a type of data; rather, it is the collection and analysis platform that connects all of the previous domains. The telemetry pipeline capability lets network operators leverage the combined universe of network monitoring data to realize greater insight into their networks and respond more quickly and intelligently to events.

The benefits that unified collection brings might be compared to the data recorder installed in modern aircraft (and, increasingly, in automobiles). Data about all key systems is consolidated into these devices — engine management, interior or cockpit, electronics, hydraulics, brakes, control and so on. In an attempt to reconstruct the cause of an accident, it is this multidimensional repository that provides analysts the best opportunity to perform diagnostics and problem identification — sometimes identifying failure conditions never anticipated in a laboratory setting. Although the stakes may not be quite as high in enterprise network management, the effects of failure in our increasingly connectivity-reliant world may be felt across a very wide area. More importantly, our ability to act on real-time analysis of this data allows us to detect and prevent failures before they happen or at least facilitate restoration and resolution.

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The use of a third-party telemetry pipeline is becoming increasingly common for managing telemetry at scale. These pipelines transform, enrich and route large volumes of data from various sources such as metrics, logs and traces. As the need for comprehensive visibility into complex systems grows, vendors such as Cribl have started to offer integration capabilities for the ingest of network telemetry as an additional dimension for troubleshooting. See Enhance Monitoring and Observability via Telemetry Pipelines for additional guidance on this topic.

Cloud service providers and cloud-based network observability solutions also transport network telemetry data so it can be used for analysis in data lakes, time series databases, event streaming platforms and other observability platforms used within an organization. For example, Kentik Firehose allows for streaming export of standardized and enriched network telemetry data into other platforms. This telemetry can be used to supplement telemetry being collected by APM-focused observability platforms used by product teams, risk and threat analytics platforms or ML/AI systems for business intelligence. Telemetry streaming can also be accomplished using cloud-based data streaming services and serverless functions.

7.1 The Network Telemetry Pipeline

No two enterprise networks are the same. Storing different facets of monitoring data in their own, possibly proprietary, tool-centric repositories, cripples the ability to diagnose complex anomalies and tackle the engineering challenges unique to its environment.

Data management and business analytics have come a long way. The success of DevOps practices has demonstrated that breaking out of traditional silos can improve responsiveness and quality. Removing the silos that separate data about the network can have a similar effect. This is the philosophy behind the telemetry pipeline, as illustrated in Figure 7.

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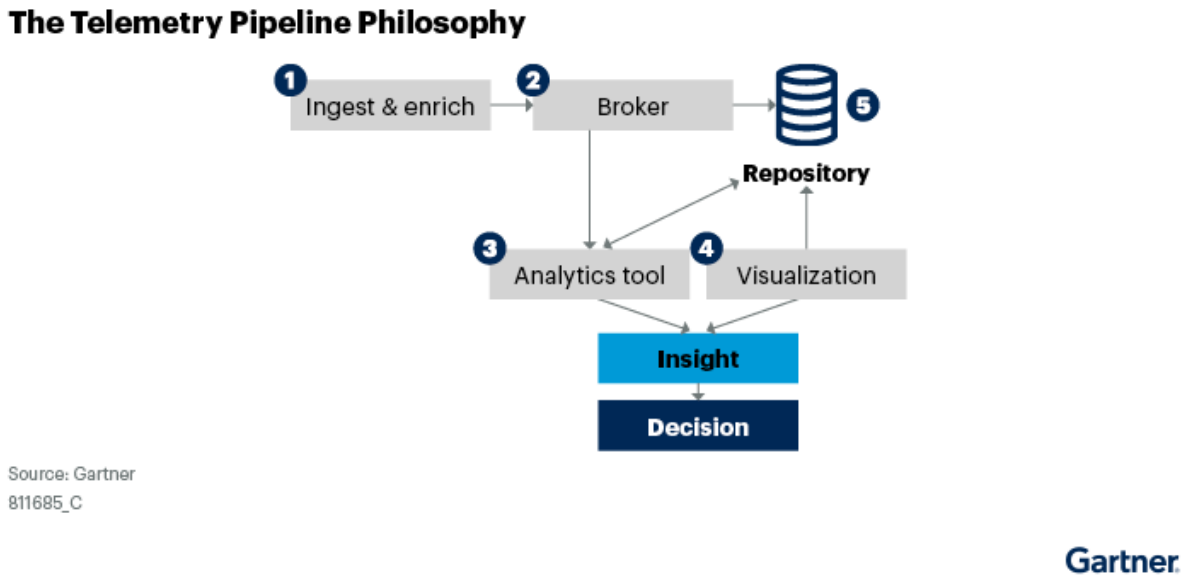
Figure 7: The Monitoring Data Pipeline



A Note on "Integration"

How systems and data can be integrated is limitless. Although the simplified block view in Figure 7 is conveniently abstract enough for discussion, actually deploying a singular telemetry pipeline as illustrated may not be possible. Expect something of a sliding scale through the use of software-based integration technology, APIs and common data formats. As long as data from different network telemetry domains can be integrated at the analytics and visualization layers, it is not problematic for the data to reside in separate repositories. Choose monitoring tools that support multiple data integration architectures through mechanisms, such as standard data formats and documented REST APIs, to obtain maximum flexibility.

7.2 The Telemetry Pipeline Components

A telemetry pipeline comprises several components, as illustrated in Figure 4 and described below, with the numbers corresponding to the call-outs in Figure 7:

1. **Ingest and enrich:** Collects and optionally transforms or normalizes telemetry data before passing it to either a broker or directly into the repository. Due to the diversity of data associated with the platform, the ingest software can take many forms. Examples of ingest software include Elastic (Logstash), InfluxData (Telegraf), Fluentd and Prometheus. The (optional) enrichment process enables context from other systems such as a configuration management system (CMS), incident management or even other monitoring solutions to be collected and associated with the incoming data.

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2. **Broker:** A hybrid component that may include aspects of ingest, analytics and repository capabilities. NPBs are deployed to monitor large networks without having to backhaul the more voluminous data, such as captured packets, to a central site. These can perform edge analytics on captured data, support event notification and even create flow records from captured packets.

3. **Analytics:** Delivered either as an AI or machine learning analysis feature within a monitoring tool or as a stand-alone event intelligence solution (formerly called "AIOps platform), it generates insight from the data once it reaches the platform. This may include initiating notifications/alerts, creating reports or correlating events to inform cause analysis. Algorithms used here may include ML and statistical analysis, as well as an ad hoc environment enabling engineers to explore raw data. Examples of tools in this category include BigPanda, Moogsoft, Cisco Secure Workload, Ciena (Blue Planet), ExtraHop RevealX Network Performance Monitoring, Elasticsearch and Splunk IT Service Intelligence (ITSI). See Solution Criteria for AIOps Platforms for additional guidance on selecting an AIOps platform for network telemetry correlation.

4. **Visualization:** The results of continuous or ad hoc analytics are often best presented visually. Having a tiered health and performance dashboard for a network is common (and recommended). If multiple analytics platforms or repositories are in use, the visualization tier can become the preferred integration platform. Examples of popular tools for this include Grafana Labs (Grafana), Elastic Kibana and InfluxData Chronograf. Many of the tools in the analytics category include a visualization component as well. These may not always facilitate the integration of data from multiple sources.

5. **Repository:** Building a platform to store network health and performance data can be challenging. The most current data is voluminous and requires immediate capture and analysis. Special-purpose repository technology exists for time series and event data, and remote storage on NPB can reduce the impact of packet capture on the central repositories. Intentionally plan your storage architecture for the telemetry pipeline, including the need to partition data by time or device, and aggregate or age-off data once it is no longer of operational importance. Examples of the types of repositories to consider include Elasticsearch, InfluxData (InfluxDB) and Kentik.

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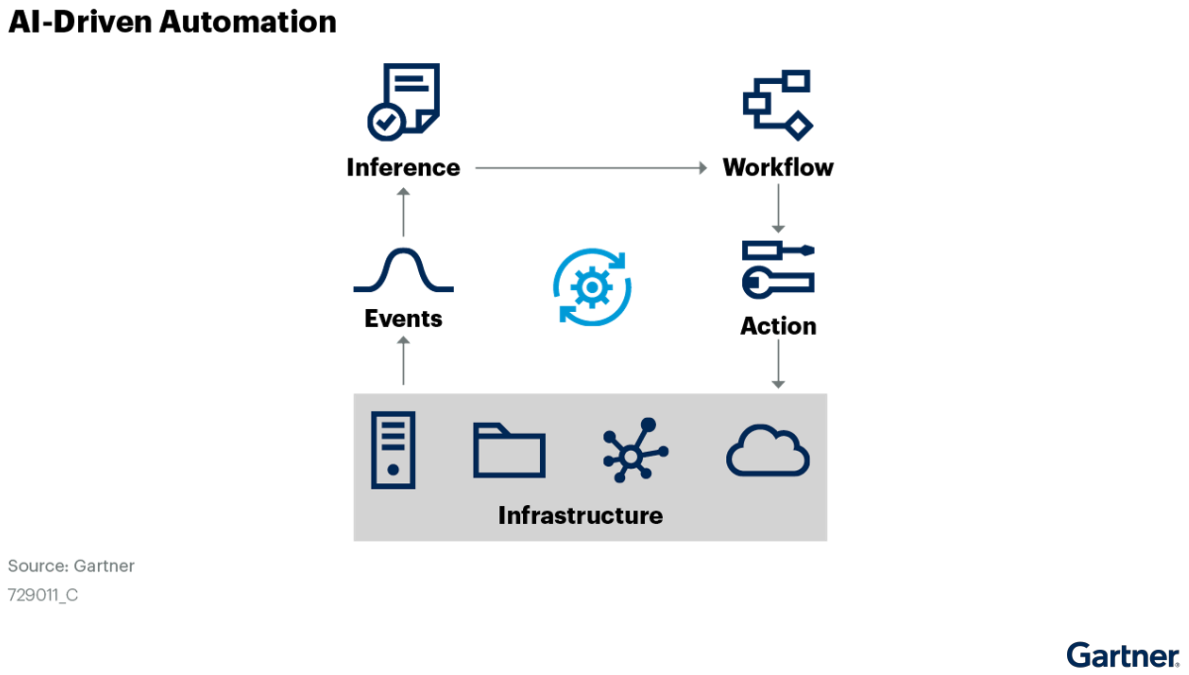
There are a variety of storage options available from public cloud providers that vary in cost, speed and complexity. Consider including them in your repository platforms, particularly for temporary or long-term purposes. Many of the vendors, products and tools listed in an individual category also apply to other categories.

Many monitoring vendors claim that they have AIOps capabilities. Gartner recognizes that some AI-driven tools can provide insight that falls below the radar of classic tools. However, not all AI use cases are mature today. The usage and maturity of event correlation, root cause analysis and capacity planning, and anomaly detection are growing rapidly. However, tools that mine general monitoring data and automatically propose and trigger automatic action, as depicted in Figure 8, are largely out of reach today.

Figure 8. AI-Driven Automation



Ask existing and prospective vendors what AIOps use cases they can address today. It is not enough for a vendor to have an AIOps strategy. When talking to a vendor about AIOps, ask them to demonstrate precise use cases that matter to you and how they can quantify the value of their solution. Figure 9 illustrates network topology, telemetry and configuration data being ingested into an AIOps platform in order to produce specific outcomes such as real-time topology analysis, performance improvement recommendations, complex incident detection and automated root cause analysis.

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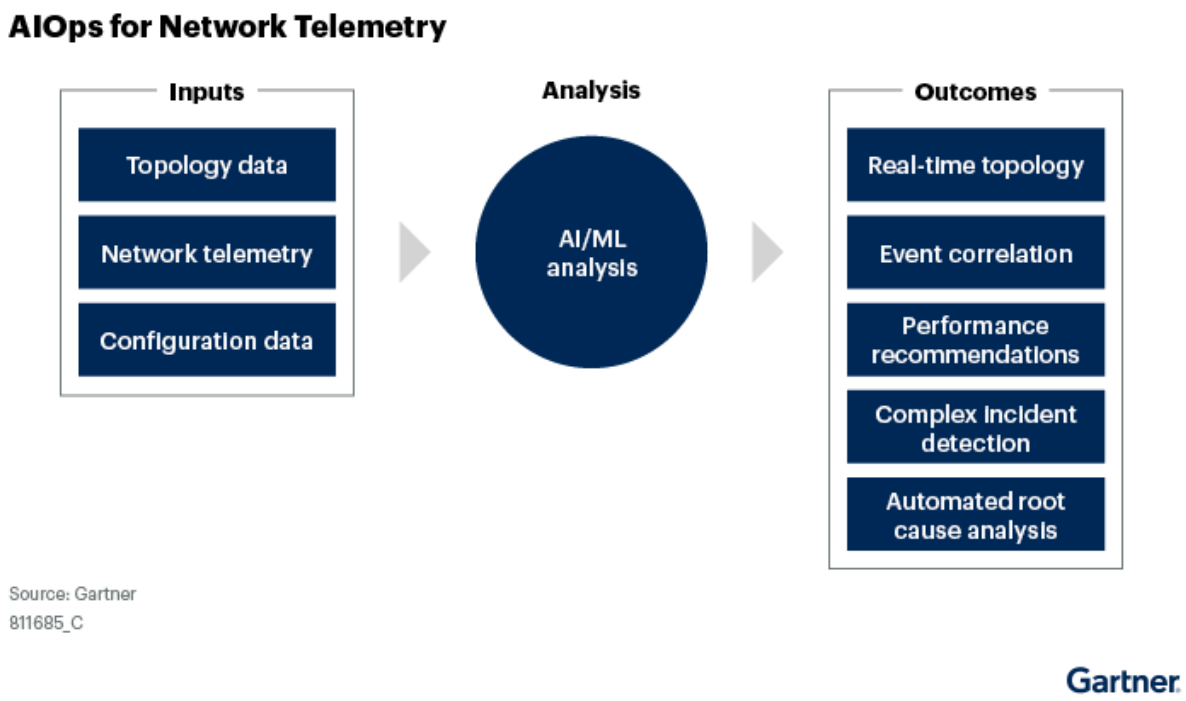
Figure 9: AIOps for Network Telemetry



7.3 Guidance

Gartner recommends that technical professionals:

- Strive for a comprehensive telemetry pipeline architecture, but leverage an iterative approach to its deployment. Prove the approach works to solve real problems, starting with a single domain and expanding the platform's capabilities from there.
- Avoid storing multiple copies of the same dataset. One goal of the telemetry pipeline is efficiency in both resource consumption and action.
- Incorporate compatibility with the telemetry pipeline collection and ingestion capabilities into product selection and base configuration when refreshing network elements.
- Assess the components used to construct the telemetry pipeline platform on a similar refresh cycle as the rest of their network elements. Advances in data management and analysis are accelerating, and better insights or a reduction in MTTR should result.

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Risks and Pitfalls

The framework, with its focus on data instead of tools, has the potential to exacerbate the number of monitoring tools organizations deploy and manage. Ultimately, the data can be aggregated in a limited number of repositories; however, in the interim, each tool might have its own repository.

Risks

- **Drowning in extraneous data:** Resist the urge to collect everything. Being selective about the signals that you collect improves focus.
- **Not collecting enough data:** A significant amount of data is necessary to get a clear picture of your network environment.
- **Not planning for data retention and expiring the data too soon:** Make sure you have enough storage to keep full data for the required amount of time. Create an archival tier where you only keep aggregated and summarized data.

Pitfalls

- **Not integrating with the rest of the organization's monitoring data:** Not only will you duplicate effort, but you will not be able to have access to the non-network event or give network context to application events.
- **Collecting too much data:** Do not collect data that does not integrate with the applications and other IT silos and is irrelevant for the business.
- **Not defining a master for a particular type of data:** Having multiple tools collecting duplicated data can result in inconsistencies that result in an incorrect conclusion, waste precious troubleshooting time and delay resolution. Define a hierarchy for every data type. In general, the packet should be authoritative when it is available.
- **Not paying attention to the internet and other shared infrastructure, such as IaaS and SaaS:** Ensure that you monitor networks that you don't own, even if your data extraction technology might not apply. Leverage synthetic transaction tools, which are often agent-based, or the cloud provider's native tooling.

Document Revision History

Transform Network Monitoring and Analytics for the Modern Era - 3 February 2021

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Rethink Network Monitoring for the Cloud Era - 6 December 2018

Recommended by the Authors

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Solution Path for Optimizing Monitoring and Observability

Monitoring and Observability for Infrastructure and Applications

Comparing Cloud Operations Approaches

Innovation Insight for Observability

A Guidance Framework for Selecting Cloud Management Tools

How to Build an Ideal Performance Monitoring Dashboard

Use Synthetic Monitoring to Enhance User Experience for Hosted and SaaS Applications

Guidance Framework for Deploying Centralized Log Monitoring

Market Guide for Digital Experience Monitoring

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Table 1: Data Type and Collection Methods

Domain	Data type	Purpose	How to collect the data
1	Log/Event Data	Awareness of changes and events to network devices and cloud components providing network services	SNMP traps, syslog, VPC Flow Logs
2	Metric Data	Collecting performance and use over time	SNMP polling, streaming telemetry, API retrieval, synthetic transactions, endpoint agents
3	Flow Data	Collecting metadata on end-to-end conversations in the network	NetFlow, sFlow and IPFIX
4	Packet Data	Collecting full conversation content for payload diagnostics and security use cases	Port mirroring, port spanning technologies, network packet brokers, taps
5	Configuration Data	Capturing device configuration change management	Screen scraping, NETCONF/yang, API, Ansible, network source of truth (NSoT) repositories
6	Forwarding, Route and Path Data	Optimizing traffic flow between endpoints	Network trace, API, polling, passive route collection

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Table 2: Representative Time Series Data Management Solutions

Solution	Vendor	Open Source	Delivery Model
Grafana Mimir	Grafana Labs	Yes	SaaS, self-hosted, open-source software (OSS)
InfluxDB	InfluxData	Yes	SaaS, self-hosted, OSS
M3	Chronosphere	Yes	SaaS, OSS
VictoriaMetrics	VictoriaMetrics	Yes	SaaS, self-hosted, OSS
TimescaleDB	Timescale	Yes	SaaS, OSS
Amazon Timestream	Amazon Web Services	No	SaaS
Time Series Insights	Microsoft Azure	No	SaaS
Timeseries Insights API	Google Cloud	No	SaaS

Source: Gartner

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